

Analysis PhD Exam, May 2009

1. State and prove the Fubini theorems.
2. This problem has three parts.
 - (a) State the Vitali integral convergence theorem.
 - (b) State the Lebesgue Dominated Convergence Theorem.
 - (c) Use (a) to prove (b) [the convergence in measure version].
3. This problem has two parts.
 - (a) Let $\mu : \Sigma \rightarrow [-\infty, \infty]$ be countably additive, where Σ is a σ -algebra. Define positive and negative sets for μ and state the Hahn decomposition theorem for μ .
 - (b) Let μ be as above. Does there exist a set $E \in \Sigma$ such that $\mu(E)$ is the minimum value for μ on Σ ? Prove.
4. Suppose $\{f_n\}$ is a sequence of integrable functions with the property that $\sum \int |f_n| d\mu < \infty$. Analyze the pointwise convergence of $\sum f_n(x)$, for $x \in X$.
5. Let Y be an orthonormal set in a Hilbert space and suppose $f : Y \rightarrow \mathbb{C}$ is given. State necessary and sufficient conditions for $\sum_{y \in Y} f(y)y$ to converge. Prove.
6. Let X be a normed space and $\pi : X \rightarrow X^{**}$ the natural map. Under what conditions is $\pi(X)$ closed in X^{**} (equipped with the norm topology)? Prove.
7. This problem has two parts.
 - (a) State the Principle of Uniform Boundedness for a family of operators.
 - (b) Let X be a normed space. Use (a) to show that if $A \subset X$ satisfies $\sup_{a \in A} |x^*a| < \infty$ for each $x^* \in X^*$, then $\sup_{a \in A} \|a\| < \infty$.
8. Let X be a normed space, and assume M is a closed linear subspace of X . Let $z \in X \setminus M$ and $S = \text{span}\{z, M\}$. Show S is closed in X . [Hint: Let $f : S \rightarrow \mathbb{C}$ be defined by $f(x + \alpha z) = \alpha$, for $x \in M$ and $\alpha \in \mathbb{C}$. Show $\|f\| \leq 1/\text{dist}(z, M)$, hence $f \in S^*$.]